

TSMC

CoWoS and Advanced Packaging Updates

- Nudging up CoWoS capacity further in 2027 and 2028:** We are raising our CoWoS capacity estimate for the industry by 4%/6% in 2027/28, led by a minor expansion at TSMC (mostly incremental CoWoS-R capacity) and stronger expansion at OSAT vendors (primarily for CPU projects). We now expect TSMC's CoWoS capacity to reach 200K/225K wfpw by end 2027/28 with around 10k wfpw of additional capacity expansion for CoWoS-R to fulfill Vera CPU and AI ASIC demand. The faster expansion is at OSATs, whose combined capacity is likely to reach 60K/95K wfpw by end 2027/28, primarily focused on CPUs and expansion of CoWoS-R and similar applications. CoWoS demand-supply gap is now in the 10% range, and has shrunk a bit recently, with leading edge wafers and substrate becoming the bigger bottlenecks for AI customers.
- SoIC activity picking up in 2028, with NVDA Feynman, more designs to come in 2029:** 3D SoIC roadmap is becoming clearer, in our view, with NVDA Feynman likely to adopt A16 on A16 logic-logic SoIC stacking for the compute dies. There are several other ASICs (MTIA 600, OpenAI Serrano, potentially one version of Google TPUv10 or similar family of chips) which are also likely to adopt SoIC. We believe that at N2 and beyond, most AI accelerators will either need to choose 3D SoIC or very large reticle size 2.5D packages (like EMIB) to accommodate the compute chiplets, I/O and HBM cubes. We expect TSMC to speed up SoIC capacity expansion from 2H27, reaching 35K wfpw by end 2027 and 70K wfpw by end 2028, with the 2028 estimates slightly higher than our prior expectations. Converting SoIC capacity to chip output is not as straightforward as CoWoS, given the wide variance in bottom-die specifications and yield - our current back-of-envelope calculation indicates that it would require ~10k wfpw of SoIC capacity to support 1M accelerators (each containing 2 reticle size base dies, similar to AMD MI 350 or 450). As a result, we expect SoIC capacity expansion to remain quite strong beyond 2028.
- NVDA estimates slightly higher driven by Rubin Ultra in 2027 and Feynman in 2028:** We are nudging up NVDA CoWoS estimates slightly again (4%/4% for 2027/28) to reflect a stronger ramp for Rubin and Rubin Ultra in 2027 and Feynman in 2028. We believe that the upswing in demand from Neoclouds and SPCX is creating more upside for Rubin/Rubin Ultra in 2027. We also believe that the cross-over to Rubin Ultra may happen quicker than anticipated, due to limited design spec changes between these two chips. Looking into 2028, we expect Feynman to now adopt a 3D SoIC packaging for logic-logic integration (our current view is that NVDA may use four dies in a two-stack configuration, with each stack having two A16 logic compute dies stacked on top of each other). I/O and other circuitry may be on separate dies, similar to Rubin. This is likely to limit the 2.5D package size for Feynman to 6-7x reticle limit on CoWoS-L, while still accommodating 4 compute dies and 12 to 16 HBM cubes within the same package. In addition to GPUs, we continue to see strong demand for Vera CPUs, with TSMC expanding CoWoS-R capacity and additional support from Amkor (in 2H26/2027) and ASE (late 2027). We also expect LP40 to migrate to CoWoS-R in late 2027, while

Overweight

2330.TW, 2330 TT

Price (17 Sep 26): NT\$2425.0

Price Target (Jun-27): NT\$3200.0

Technology and Telecoms

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Quarterly Forecasts (FYE Dec)

Adj. EPS (NT\$)

	2025A	2026E	2027E
Q1	13.94	22.08A	31.56
Q2	15.36	27.25A	32.82
Q3	17.44	27.72	37.71
Q4	19.50	31.10	39.28
FY	66.24	108.15	141.37

Style Exposure

Quant Factors	Current	Hist %Rank (1=Top)			
	%Rank	6M	1Y	3Y	5Y
Value	38	46	39	39	76
Growth	35	41	16	24	11
Momentum	40	23	25	49	69
Quality	12	29	10	12	1
Low Vol	26	30	32	37	26
ESGQ	20	90	20	10	4

Sources for: Style Exposure – J.P. Morgan Global Markets Strategy; all other tables are company data and J.P. Morgan estimates.

See page 12 for analyst certification and important disclosures, including non-US analyst disclosures.

Price Performance



	YTD	1m	3m	12m
Abs	56.4%	1.0%	1.7%	91.7%
Rel	-3.4%	0.1%	0.8%	9.7%

Company Data

Shares O/S (mn)	25,930
52-week range (NT\$)	2535.0-1265.0
Market cap (\$ mn)	1,977,022
Exchange rate	31.81
Free float (%)	92.1%
3M ADV (mn)	31.95
3M ADV (\$ mn)	2,404.0
Volatility (90 Day)	34
Index	TAIEX
BBG ANR (Buy Hold Sell)	39 2 0

Key Metrics (FYE Dec)

NT\$ in millions	FY25A	FY26E	FY27E	FY28E
Financial Estimates				
Revenue	3,809,054	5,479,260	7,443,533	9,319,830
Adj. EBIT	1,936,092	3,206,593	4,365,825	5,564,701
Adj. EBITDA	2,624,188	4,014,720	5,334,428	6,833,311
Adj. net income	1,717,883	2,804,474	3,666,124	4,668,019
Adj. EPS	66.24	108.15	141.37	180.01
BBG EPS	64.73	107.38	140.98	176.64
Cashflow from operations	2,437,716	3,564,030	4,453,920	5,822,668
FCFF	1,292,759	2,201,778	1,861,920	3,230,668
Margins and Growth				
Revenue Growth Y/Y (%)	31.6%	43.8%	35.8%	25.2%
EBIT margin	50.8%	58.5%	58.7%	59.7%
EBIT Growth Y/Y (%)	46.4%	65.6%	36.2%	27.5%
EBITDA margin	68.9%	73.3%	71.7%	73.3%
EBITDA Growth Y/Y (%)	32.2%	53.0%	32.9%	28.1%
Net margin	45.1%	51.2%	49.3%	50.1%
Adj. EPS growth	46.4%	63.3%	30.7%	27.3%
Ratios				
Adj. tax rate	16.0%	17.2%	17.7%	17.7%
Interest cover	NM	NM	NM	NM
Net debt/Equity	NM	NM	NM	NM
Net debt/EBITDA	NM	NM	NM	NM
ROE	35.4%	42.7%	40.3%	38.0%
Valuation				
FCFF yield	2.1%	3.5%	3.0%	5.1%
Dividend yield	0.9%	1.2%	1.4%	1.7%
EV/Revenue	16.3	11.0	8.0	6.2
EV/EBITDA	23.6	15.0	11.1	8.4
Adj. P/E	36.6	22.4	17.2	13.5

Summary Investment Thesis and Valuation

Investment Thesis

We are OW on TSMC, given the increased confidence in AI demand, along with an aggressive capex plan ahead. We believe AI demand visibility is extended to 2029/30 due to the solid accelerator product line-up in the coming years. In response, the company is also accelerating its capacity plan further in order to capture incremental AI demand and leave limited room for potential competition. We believe TSMC's leading industry position is intact with its leading-edge process technology remaining ahead of its peers and its mature nodes continuing to grow with a focus on high value segments. For GMs, we believe it should stay on a slowly upward sloping curve once N2 dilution winds down, due to price increases, better operating efficiency and continued high utilization.

Valuation

Our Jun-27 PT of NT\$3,200 is based on ~20x 12-month forward P/E and reflects strong revenue growth from stronger AI demand and faster capacity growth. Our target multiple is higher than TSMC's five-year average historical multiple.

Performance Drivers

Market	72%
Region	20%
Macro	1%
Style	2%
Idiosyn.	6%

Factors	6M Corr	1Y Corr
Market: MSCI Asia Pac ex JP	0.90	0.85
Region: Taiwan	0.81	0.85
Macro:		
JPM EM Currency(EMCI) Fixing	-0.32	-0.22
JP Morgan GBI-EM Global Div	-0.21	-0.17
Citi Economic Surprise - EM	0.06	0.15
Quant Styles:		
Size	-0.23	-0.18
Momentum	-0.04	0.07
DivYld	0.04	-0.06

Spectrum X CPO is also likely to become a meaningful contributor to CoWoS demand in 2H27. In 2028, a key area of upside would come from NVDA deciding to introduce NVLink Switch CPO, which may require significant CoWoS capacity support - at this point, we are not modeling this into our estimates, given uncertainties regarding TSMC COUPE ramp for Scale-up CPO.

- **AMD forecasts remain unchanged for MI series, continued upside for Venice despite ramp concerns:** Our estimates for AMD's MI450/500 series are largely unchanged for 2027/28 with the initial volume ramp for MI450 likely coming through in 4Q26. We expect demand to pick up significantly in 2027, gated by the ability to ramp up rack-level production for the Helios rack. Chip and package level issues that had caused some concerns in 1H26 are largely behind us. For the MI500 series, we believe that AMD has scaled back its very ambitious package size plans (up to 14-15x previously) to ~9x reticle limit, given the ongoing challenges of scaling reticle limits beyond 9x using CoWoS-L. While there have been some concerns about yield challenges in Venice CPU ramp, we believe that OSATs such as ASE are likely to ramp up significantly in volume starting in 4Q26. We are raising the CoWoS allocation for Venice and expect unit shipments to now reach 3.7M/5.5M units in 2027/28, given strong CPU demand and unconstrained N2 capacity availability at TSMC. We believe that ASE is only able to supply 60-70% of the total demand for Venice from AMD, which should result in increased reliance on TSMC and some orders migrating to Amkor, and possibly PTI in 2H27.
- **Slight uptick for TPU forecasts, mostly at Mediatek in 2027; v9 ramp with EMIB-T in 2028 largely on track:** Overall, we now expect GOOG TPU units to reach 9.5M/11.4M units in 2027/28, with bulk of the upside coming from Mediatek's TPUv8 Zebrafish and v9 Humufish. We expect Mediatek's TPUv8 to ramp up to 4.1M units in 2027, with unit shipments likely to be higher than Broadcom's TPUv8i Sunfish (although Sunfish is a 2-Compute die solution). TPUv9 Humufish ramp seems largely on track, with potential tapeout in 4Q26 and production ramp in late 2027/early 2028. Market concerns have largely centered around EMIB-T yields, but our recent checks suggest that the substrate level yields are at ~50% for its leading substrate partners, while final packaging yields are reaching 90-95%, which should be sufficient to go into mass production in 12-15 months. While there have been some questions about a Plan B on CoWoS for TPUv9 for Mediatek, we believe that this is no longer active and is unlikely to move back to CoWoS at this stage of the product evolution. For TPUv10, however, all options (EMIB-T, CoWoS-L or TSMC's version of EMIB-like packaging) are open, in our view, given the project is still in the proposal stage. We expect TPUv10 to be awarded in 4Q26 (likely lion's share going to Mediatek).
- **New AI ASICs finally starting to ramp for Broadcom in 2027:** Our forecasts are now starting to incorporate the ramp of new AI ASICs at Broadcom in 2027. We see the MTIA family reaching ~0.8M units in 2027, with a bigger ramp to ~1.1M units in 2028. Most of the volume in 2027 is still Iris (MTIA 400), while Arke and Astrid (MTIA 450/500) are likely to start ramping at higher volume in 2028. We also see a pickup in OpenAI Jalapeno shipments in 2027 (we expect ~400-500K units), while the ramp in 2028 could be stronger, given that the second generation (Serrano) with more powerful compute and 3D SoIC packaging could see wider adoption. We would still need to monitor the pace of evolution, given readiness in the downstream supply chain (board level, rack level) is also essential for a successful volume ramp in 2027. At Marvell, we continue to see MSFT MAIA 300 ramp to ~0.7M units in 2027, although the MAIA 400 project award remains unclear for now.
- **Stronger Trainium 3 and 4 expectations in 2027/28, with better CoWoS-R availability:** We raise our Trainium forecasts to 3.6M/3.7M units in 2027/28 (from 3.5M/3.5M), given better CoWoS-R availability and stronger demand. Trainium 3 successfully entered mass production in late May, with volume ramping rapidly in Jun-

Aug, and we expect it to remain at a high level from 4Q26 through at least mid-2027. Trainium 4 is on track to tape out in late 2026, with potential mass production earliest in 4Q27. Based on our checks, Alchip is the sole back-end design-service partner for both Trainium 3 and Trainium 4. While other design service vendors such as Marvell could participate in Trainium 4 with an I/O die on a different interconnect solution, this would likely come at a later stage and at a smaller volume. On packaging choice, while CoWoS should remain the mainstream solution for Trainium, EMIB could account for a small portion of Trainium 3 and 4 as a diversification option.

- **EMIB-T seems ready to ramp, despite yield struggles, EMIB-like solutions emerging:** Intel EMIB-T should be on-track for a ramp in early 2028 with TPUv9 being the primary product, along with some volume for Trainium 4. Most of the value capture in the EMIB-T foodchain is with substrate vendors, with increased demand for substrate formation, TCB-based die-attach for embedded silicon bridges and lamination. OSATs will also have some role to play for wafer bumping, while final package mounting and testing should be done by Intel in-house (in AZ or Malaysia). Demand for EMIB-like solutions (RDL on substrate) is increasing from Fabless customers due to lower costs compared to CoWoS packages of equivalent reticle size (up to 50% cheaper at optimal yields). We believe TSMC is also exploring similar technologies in partnership with substrate vendors (Unimicron, Kinsus) and OSAT vendors.
- **CoPoS progress slow, and larger reticle size CoWoS projects also seeing some initial challenges:** Our checks suggest TSMC's CoPoS is progressing more slowly than expected, with risk production likely only in 2028 and mass production in 2029 at the earliest. We attribute the slowdown mainly to engineering challenges—such as warpage and glass cracking—that must be resolved before entering mass production. This trajectory is broadly consistent with what we hear from OSATs (such as ASE), whose CoPoS-like PLP solutions remain in the engineering phase and are not expected to reach mass production before 2029. TSMC is attempting to scale CoWoS-L packages to 9-10x reticle size or larger in 2027-28 (current largest package is MI450 with 5.5x reticle size). However, our recent checks suggest slightly slower progress in these efforts due to thermal and warpage challenges that need to be surmounted. Our checks indicate that AMD MI500 is likely to be the first project (scaled down from 15x reticle size previously to 9x, with a Plan B to stay at ~6x reticle size), while Feynman will scale up only to 6-7x reticle size (due to the usage of 3D SoIC). We believe TSMC's preferred approach is for customers to leverage 3D SoIC as the additional dimension for scaling, rather than trying to scale CoWoS-L to extremely large package sizes (14x reticle size will only yield 3 packages per CoWoS wafer, resulting in significant wastage of silicon). GPU customers appear to be willing to use 3D SoIC (AMD already, NVDA from Feynman), but large ASIC clients (TPU, Trainium) are still hesitant, with MTIA 600 and OpenAI Serrano likely the first SoIC-based ASIC projects in 2028-29.

Note: Alphabet (GOOGL, OW), Amazon.com (AMZN, OW), SpaceX (SPCX, OW), and Meta Platforms (META, OW) are covered by Doug Anmuth. Advanced Micro Devices (AMD, N), Broadcom (AVGO, OW), Marvell Technology (MRVL, OW), Intel (INTC, UW) and NVIDIA (NVDA, OW) are covered by Harlan Sur. Amkor Technology (AMKR, OW) is covered by Mayur Ramdhani. Apple (AAPL, OW) and Microsoft (MSFT, OW) are covered by Samik Chatterjee, CFA. Open AI is covered by Aaron Steiker.

Table 1: CoWoS allocation breakdown

CoWoS Consumption by customer (k wafer per year)	2023	2024	2025	2026E	2027E	2028E	% Revision vs prior ests			JPM Comments
							2026E	2027E	2028E	
NVIDIA	70	192	400	725	1240	1800	0%	4%	4%	Uptick in 2027 and 2028 from stronger Rubin family, and upside in 2028 from Feynman, Rosa CPU, LPX and increased CPO ramp for SpectrumX. Not factoring in NVLink Switch CPO in 2028 yet. CoPoS not ready before 2029.
At TSMC	70	127	385	675	1060	1400	-1%	-1%	1%	2026 CoWoS-L at TSMC, others at OSAT, 2027 Rubin Ultra assuming 2-die-on-interposer solution. 2028 Feynman SoIC based CoWoS-L with ~7x reticle limit.
CoWoS-S / R	70	97	30	30	50	100	200%	25%	-5%	TSMC adding some CoWoS-R to support Vera and future LPX projects now.
CoWoS-L	0	30	355	645	1010	1300	-4%	-3%	2%	Majority of CoWoS to migrate to CoWoS-L by 2H25. Rubin Ultra likely to remain on CoWoS-L. Feynman likely to be still on CoWoS-L with 7x reticle size package vs. 5.5x for Rubin / Rubin Ultra.
Outside TSMC (CoWoS-S/R)	0	65	15	50	180	400	25%	50%	14%	Outsourcing ratio picking up meaningfully from 2026 and accelerate in 2027 as Amkor and ASE / SPIL pick up more CoWoS-R share for Vera. LP40 on CoWoS-R could also be outsourced to OSATs in 2028.
AMD	10	45	52	95	380	590	0%	7%	9%	MI450 ramp in late 2026, moving to CoWoS-L for MI450 onwards, forecasts incl. Xilinx. Keeping MI series forecasts intact, but raising Venice CPU demand. Venice in ASE from 2027, some volume going to Amkor and PTI.
CoWoS-S	10	45	52	15	0	0	-40%	NA	NA	MI350 and Xilinx - main CoWoS-S customer. AMD also has non-AI projects such as PC, game console migrating to 2.5D packaging in 2H26, likely to be mostly at ASE and PTI but not captured in these forecasts.
CoWoS-L	0	0	0	55	145	230	0%	-9%	0%	CoWoS-L move likely only for MI400 family and beyond, Venice family of CPUs also to use CoWoS-L.
Outside TSMC (CoWoS-L/ FoCoS / EFB)	0	0	0	25	235	360	67%	21%	16%	Mostly Venice CPU, but several PC, Gaming and Xbox chips also using Wafer-Level Fan-Out from 2027 - we are modeling only the server CPU parts. Mostly ASE, but some Amkor as well in late 2027 and 2028.
Broadcom	40	70	96	250	420	615	0%	-1%	6%	TPU v7 (Ironwood) and v8 most of the volume in 2026 and 2027, v8 family likely to extend to 2028 as well as Sunfish Ultra. MTIA and OpenAI Jalapeno/Serrano ramp starting to show up in forecasts now.
CoWoS-S at TSMC	40	70	91	220	280	165	0%	8%	27%	CoWoS-S is still mainstream in 2026, but more adoption of CoWoS-L from 2027 for Meta, OpenAI and other ASIC projects.
CoWoS-S outside TSMC	0	0	5	10	50	130	0%	-29%	0%	Outsourcing starting to pick up - trialling ASE and Amkor for Networking projects in 2027. OSAT qualification slower than expected. Inhouse packaging at Singapore (with Toppa) unlikely to be a large portion into 2028.
CoWoS-L	0	0	0	20	90	320	0%	-5%	0%	Starting to use CoWoS-L for some projects; also likely to incorporate SoIC for some advanced ASICs such as MTIA 450/500 and OpenAI Titan2 in 2028.
Alchip + AWS	7	14	5	80	180	265	23%	3%	6%	Trn2 upside in 1H26, Trn3 ramp in mid 2026, upside to lifetime volumes for Trn3. Most Trn3 CoWoS allocation through AWS or Alchip. Small overflow to other vendors like Marvell due to excess demand.
Marvell (incl AWS for Trn2)	2	13	75	35	40	80	0%	0%	0%	Near-term upside to Trn2. Minor Trn3 CoWoS or OSAT allocation. MSFT MAIA 3 to ramp in 2027. Not seeing a big ramp for other CoWoS projects in 2027. In 2028, expect some allocation for Trn4 (for NVLink fusion versions).
MediaTek	0	0	0	30	171	55	0%	14%	-27%	2026 TPUv8t volume strengthening in 2026/27, with upside in 2027 due to better substrate supply. TPUv9 Humulish now firmly on Intel EMIB-T - hence lower 2028 CoWoS allocation.
Others	5	16	24	35	40	50	-13%	0%	-17%	Includes Cisco, Microsoft, GLIC, Socionext, other ASIC design houses and smaller startup customers etc.; some non-AI chips also using CoWoS (Networking chips for various vendors, some small volume server CPUs etc.).
Overall Total CoWoS Consumption yoy	134	350	652	1,250	2,471	3,455	1%	4%	4%	
		161%	86%	92%	98%	40%				
CoWoS Consumption by customer (k wafer per year)	2023	2024	2025	2026E	2027E	2028E	2026E	2027E	2028E	JPM Comments
Total TSMC	134	285	632	1,165	2,006	2,565	2%	3%	3%	TSMC includes sub-contracting of oS to ASE/SPIL from 2025 and sub-contracting of Interposer to Vanguard.
Total Others (OSATs)	0	65	20	85	465	890	-15%	9%	7%	Mainly ASE/SPIL and Amkor, more full-stack CoWoS from 2H26. Intel EMIB not modeled here, but main volume ramp only in 2028.
Overall Total CoWoS Consumption	134	350	652	1250	2471	3455	1%	4%	4%	

Source: Company data, J.P. Morgan estimates. Note: Capacity allocation is before adjusting for yield losses, which may vary by accelerator and package type.

Table 2: WMCM forecasts

WMCM Consumption (K wafer per year)	2023	2024	2025	2026E	2027E	2028E	% Revision vs prior ests			JPM Comments
							2026E	2027E	2028E	
Apple	0	0	0	417	720	1,044	0%	0%	0%	All high-end iPhone 18 models in 2H26E using WMCM.
Number of iPhones (M units)				88	151	209	0%	0%	-5%	Likely to adopt for A20 Pro based models on N2. Encompassing iPhone foldable, Pro and Pro max.
Utilization				100%	100%	100%	0%	0%	0%	
Gross Die per wafer				210	210	200	0%	0%	-5%	Expect Package size to be 50% larger than inFO with ~210 dies per Wafer due to side-by-side stacking of LPDDR5.
% Yield				98%	98%	98%	0%	0%	0%	
TSMC WMCM Capacity (k wafer per year)				426	735	1,065	0%	0%	0%	

Source: Company data, J.P. Morgan estimates.

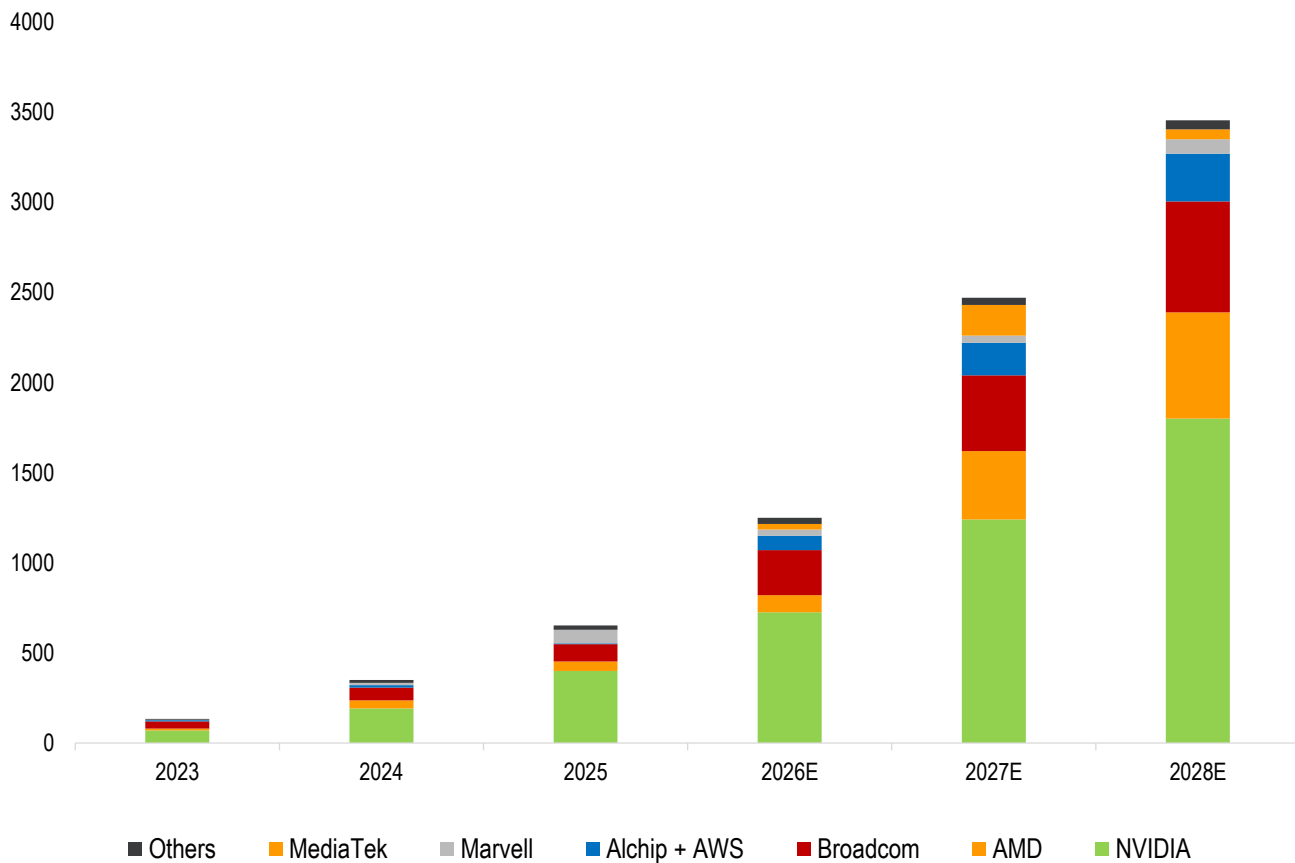
Table 3: GPU and AI accelerators chip-level shipment forecasts

GPU and Accelerator Units (M)	2023	2024	2025	2026E	2027E	2028E	% Revision vs prior ests		
							2026E	2027E	2028E
NVDA									
Hopper Series	1.7	4.3	1.0	0.4	-	-	0%	NA	NA
Blackwell Series	-	0.2	5.2	6.4	1.0	-	0%	0%	NA
Rubin Series	-	-	-	2.1	5.3	1.0	0%	-27%	-33%
Rubin Ultra Series (2 die equivalent)	-	-	-	-	4.0	8.7	NA	135%	2%
Feynman Series	-	-	-	-	-	2.5	NA	NA	53%
Total NVDA GPU Units	1.7	4.5	6.2	8.9	10.3	12.2	0%	3%	5%
NVDA eq GPU die units	1.7	4.7	11.4	17.4	20.5	29.4	0%	3%	11%
Vera CPU	-	-	-	0.7	5.0	3.0	17%	0%	0%
Rosa CPU	-	-	-	-	-	4.0	NA	NA	0%
Others Including Spectrum X	-	-	-	0.2	0.5	0.9	5%	43%	0%
LPX (LP40)	-	-	-	-	-	1.0	NA	NA	0%
AMD									
MI300/325	0.1	0.5	0.4	-	-	-	NA	NA	NA
MI350	-	-	0.3	0.2	-	-	0%	NA	NA
MI450	-	-	-	0.5	1.5	1.8	0%	0%	-11%
MI500	-	-	-	-	-	0.5	NA	NA	82%
Total AMD GPU Units	0.1	0.5	0.7	0.7	1.5	2.3	0%	0%	0%
Venice CPU	-	-	-	0.4	3.7	5.5	0%	16%	22%
Google TPUs									
v4	0.7	0.4	-	-	-	-	NA	NA	NA
v5	0.5	1.3	-	-	-	-	NA	NA	NA
v6 /v7 (Trillium/Ironwood)	-	-	1.7	2.7	1.7	-	0%	0%	NA
v8t - Zebrafish - Mediatek (1 die solution)	-	-	-	0.7	4.1	1.3	40%	37%	-12%
v8i - Sunfish /Sunfish Ultra (in 2 die equivalent)	-	-	0.1	1.3	3.7	6.7	5%	-11%	-5%
v9 - Humufish - Mediatek (4 die solution)	-	-	-	-	-	3.3	NA	NA	10%
Total TPU Units	1.2	1.7	1.8	4.6	9.5	11.4	6%	7%	-2%
Total TPU Compute Dies	1.2	1.7	1.8	8.6	14.9	28.0	4%	1%	1%
AWS									
Inferentia 2	0.3	0.5	0.1	-	-	-	NA	NA	NA
Trainium 2	-	0.3	1.5	0.6	-	-	0%	NA	NA
Trainium 3	-	-	-	1.6	3.6	-	23%	3%	NA
Trainium 4	-	-	-	-	-	3.7	NA	NA	5%
Total Inferentia / Trainium Units	0.3	0.8	1.5	2.2	3.6	3.7	16%	3%	5%
Total Trainium / Inferentia Compute dies	0.3	1.0	3.0	4.4	7.2	14.6	16%	3%	5%
Meta									
MTIA 300/400/450/500	-	-	-	0.2	0.8	1.1	-20%	52%	0%
Total MTIA units (with CoWoS)	-	-	-	0.2	0.8	1.1	-20%	52%	0%
Microsoft									
MAIA 200 - GUC	-	-	-	0.1	0.1	0.0	0%	0%	NA
MAIA 300 - Marvell	-	-	-	0.1	0.7	0.8	0%	0%	0%
Total MAIA units	-	-	-	0.2	0.8	0.8	0%	0%	0%

Source: Company data, J.P. Morgan estimates.

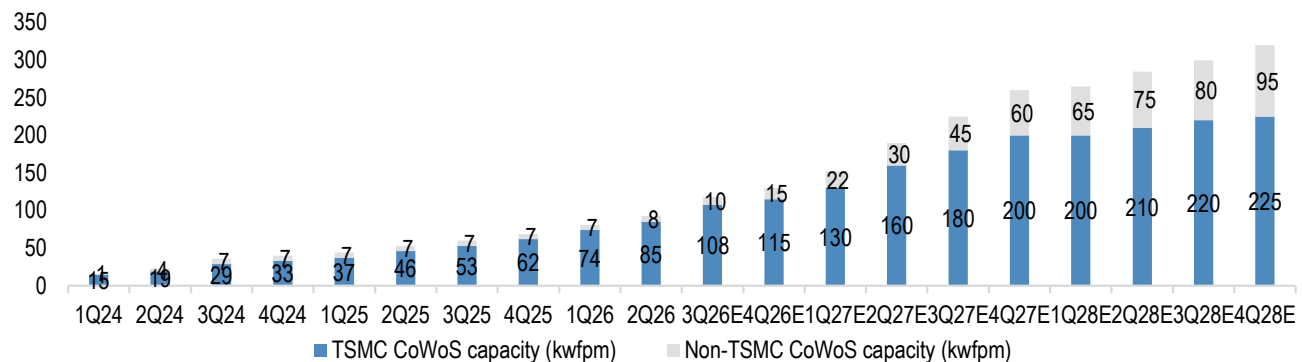
Figure 1: Customer mix for CoWoS

k wafers per year



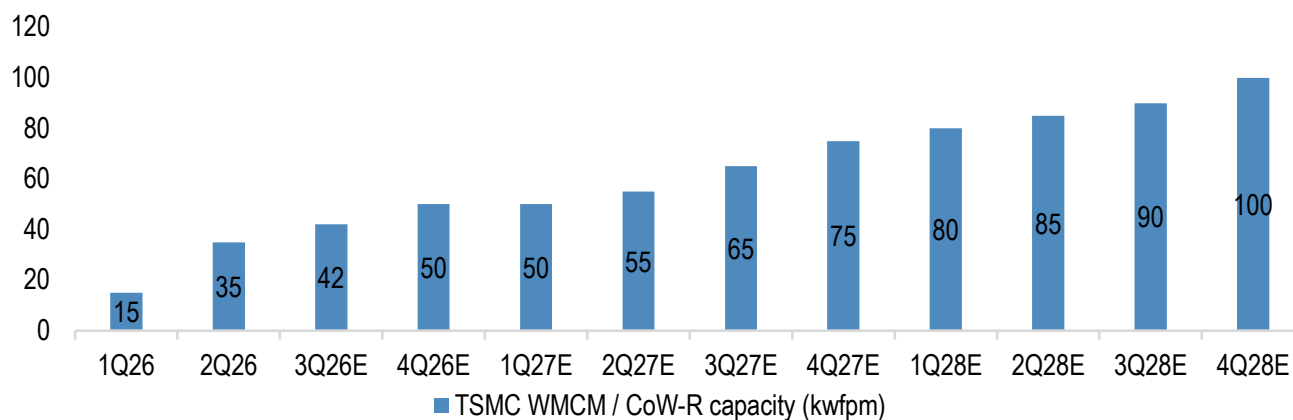
Source: Company data, J.P. Morgan estimates.

Figure 2: CoWoS capacity – TSMC/non-TSMC



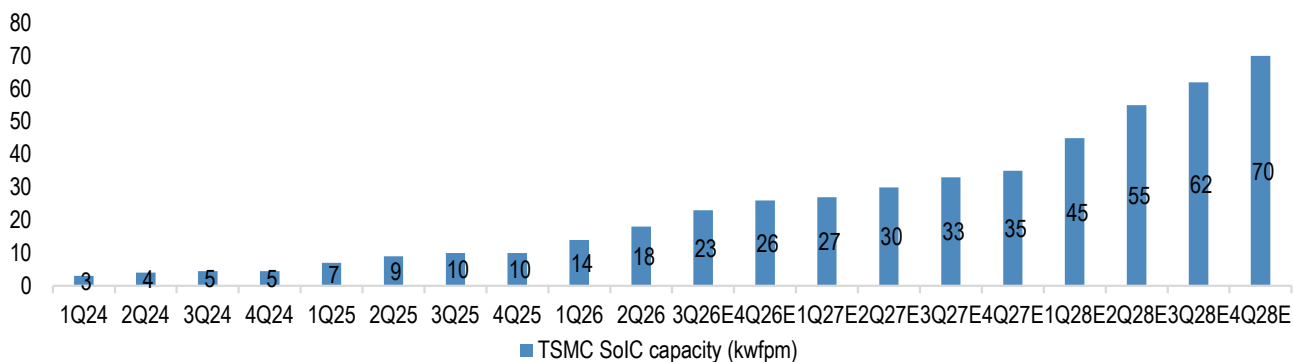
Source: Company data, J.P. Morgan estimates.

Figure 3: WMCM/CoW-R capacity – TSMC



Source: J.P. Morgan estimates.

Figure 4: SolC capacity – TSMC



Source: Company data, J.P. Morgan estimates.

Table 4: Glossary of important terms

HBM	High Bandwidth Memory	HBM is a high-speed computer memory interface for 3D-stacked synchronous dynamic random-access memory (SDRAM) from Samsung, Micron and SK Hynix.
CoWoS	Chips-on-Wafer-on-Substrate	TSMC's 2.5D interposer-based packaging technology for high-performance computing applications.
CoWoS - S	Chips-on-Wafer-on-Substrate with Silicon Interposer	A CoWoS technique using silicon interposer as the bridge between chip and substrate.
CoWoS - R	Chips-on-Wafer-on-Substrate with Redistribution Layer Interposer	A CoWoS technique using redistribution layer interposer (RDL) to replace silicon interposer, enabling higher packaging size flexibility and lower cost.
CoWoS - L	Chips-on-Wafer-on-Substrate with Local Silicon Interconnect	A CoWoS technique using local silicon interconnect (LSI) to connect dies at high speed rate, while the other area on the chip will use RDL.
COW - R	Chip-on-Wafer with Redistribution Layer Interposer	The first part of the CoWoS process, where the logic chip and HBMs are attached to the RDL interposer.
RDL	Redistribution Layer	A layer of metal interconnects that redistributes I/O access across the chip, facilitating flexible and efficient chip-to-chip connections, particularly in advanced packaging techniques like 2.5D and 3D integration.
WMCM	Wafer-level Multichip Module	An approach that integrates multiple chips (CPU, GPU, memory, etc.) at the wafer stage before slicing them into individual modules. Unlike traditional InFO (Integrated Fan-Out) packaging, focusing on single-die integration, WMCM enables more flexibility in combining different components while maintaining a small footprint.
XPU	X processing unit	The "X" in "XPU" stands for any compute architecture (CPU, GPU, ASIC etc.) that is suitable for different applications.
TPU	Tensor processing unit	Google Cloud TPUs are customized AI accelerators, optimized for training and inference of AI models.
LPDDR	Low Power Double Data Rate	Also known as LPDRAM, LPDDR is used in devices requiring speedy, high-bandwidth data rates without compromising power efficiency.
CPO	Co-Packaged Optics	An integration of optical components and ASICs on a single packaged substrate, enabling high-speed data transmission and less power loss.
ASIC	Application-Specific Integrated Circuit	ASICs (application-specific integrated circuits) are customized chips for certain special purposes, and right-of-use is limited to the chip designer only.
CSP	Cloud Service Provider	A third-party company that provides a cloud-based platform for businesses to access on demand over a network, including compute, storage, and infrastructure services.
FOCoS	Fan Out Chip on Substrate (similar to TSMC's CoWoS)	A fan-out package flip-chip placed on a high pin count ball grid array (BGA) substrate. The fan-out package has an RDL allowing the construction of shorter die-to-die (D2D) interconnections between chips.
Trn	Trainium	Trainium are chips developed by AWS specifically for AI training and inference, which deliver high performance while saving costs.
InFO	Integrated Fan-Out	A wafer-level integration technology features high-density RDL and Through InFO Via (TIV) for enhanced interconnect performance. It supports various 2D and 3D package schemes optimized for mobile and high-performance computing applications.
EIC-PIC	Electronic/Photonic Integrated Circuit	EICs utilize electronic components (transistors, capacitors) for electrical signal processing, while PICs use optical components (waveguides, lasers) for light manipulation in data transmission. EICs handle electrons, and PICs handle photons.
CoPoS	Chip-on-Panel-on-Substrate	TSMC's CoPoS is a rectangular substrate evolution of CoWoS, offering more usable space than traditional round wafers, enhancing output efficiency and reducing costs.
EMIB	Embedded Multi-die Interconnect Bridge	Intel's EMIB is an advanced packaging technology that embeds silicon bridges into an organic substrate. It enables multiple chiplets to communicate side-by-side with high bandwidth and low latency, without the need for full-size silicon interposers.

Source: J.P. Morgan.

Investment Thesis, Valuation and Risks

TSMC (*Overweight; Price Target: NT\$3200.0*)

Investment Thesis

We are OW on TSMC, given the increased confidence in AI demand, along with an aggressive capex plan ahead. We believe AI demand visibility is extended to 2029/30 due to the solid accelerator product line-up in the coming years. In response, the company is also accelerating its capacity plan further in order to capture incremental AI demand and leave limited room for potential competition. We believe TSMC's leading industry position is intact with its leading-edge process technology remaining ahead of its peers and its mature nodes continuing to grow with a focus on high value segments. For GMs, we believe it should stay on a slowly upward sloping curve once N2 dilution winds down, due to price increases, better operating efficiency and continued high utilization.

Valuation

Our Jun-27 PT of NT\$3,200 is based on ~20x 12-month forward P/E and reflects strong revenue growth from stronger AI demand and faster capacity growth. Our target multiple is higher than TSMC's five-year average historical multiple.

Risks to Rating and Price Target

Key downside risks to our rating and price target include: (1) the debate on the duration of the AI capex growth cycle and (2) any negative impact from weak PC/Smartphone demand in 2H26.

TSMC: Summary of Financials

Income Statement - Annual					Income Statement - Quarterly				
	FY24A	FY25A	FY26E	FY27E		1Q26A	2Q26A	3Q26E	4Q26E
Revenue	2,894,308	3,809,054	5,479,260	7,443,533	Revenue	1,134,103	1,270,381	1,456,995	1,617,780
COGS	(1,269,954)	(1,527,760)	(1,816,049)	(2,422,677)	COGS	(382,808)	(410,070)	(488,005)	(535,165)
Gross profit	1,624,354	2,281,294	3,663,211	5,020,856	Gross profit	751,295	860,311	968,990	1,082,615
SG&A	(98,119)	(98,775)	(122,004)	(186,088)	SG&A	(24,573)	(20,562)	(36,425)	(40,445)
Adj. EBITDA	1,984,850	2,624,188	4,014,720	5,334,428	Adj. EBITDA	824,417	965,141	1,060,120	1,165,042
D&A	(662,797)	(688,096)	(808,126)	(968,602)	D&A	(165,450)	(198,538)	(219,346)	(224,792)
Adj. EBIT	1,322,053	1,936,092	3,206,593	4,365,825	Adj. EBIT	658,966	766,603	840,774	940,250
Net Interest	76,718	93,369	107,620	84,881	Net Interest	26,145	26,959	24,133	30,383
Adj. PBT	1,405,839	2,041,663	3,388,628	4,456,530	Adj. PBT	687,800	862,430	866,332	972,067
Tax	(233,407)	(326,266)	(583,175)	(789,530)	Tax	(114,998)	(155,649)	(147,276)	(165,251)
Minority Interest	836	2,486	(979)	(876)	Minority Interest	(322)	(219)	(219)	(219)
Adj. Net Income	1,173,268	1,717,883	2,804,474	3,666,124	Adj. Net Income	572,480	706,562	718,836	806,597
Reported EPS	45.24	66.24	108.15	141.37	Reported EPS	22.08	27.25	27.72	31.10
Adj. EPS	45.24	66.24	108.15	141.37	Adj. EPS	22.08	27.25	27.72	31.10
DPS	16.50	22.00	29.00	34.50					
Payout ratio	51.0%	48.6%	43.8%	31.9%					
Shares outstanding	25,933	25,933	25,932	25,932	Shares outstanding	25,933	25,932	25,932	25,932
Balance Sheet & Cash Flow Statement					Ratio Analysis				
	FY24A	FY25A	FY26E	FY27E		FY24A	FY25A	FY26E	FY27E
Cash and cash equivalents	2,422,020	3,068,595	4,713,285	5,635,604	Gross margin	56.1%	59.9%	66.9%	67.5%
Accounts receivable	272,088	282,059	443,227	558,398	EBITDA margin	68.6%	68.9%	73.3%	71.7%
Inventories	287,869	288,109	443,227	558,398	EBIT margin	45.7%	50.8%	58.5%	58.7%
Other current assets	106,376	178,367	143,803	181,169	Net profit margin	40.5%	45.1%	51.2%	49.3%
Current assets	3,088,352	3,817,131	5,743,542	6,933,569	ROE	30.3%	35.4%	42.7%	40.3%
PP&E	3,234,980	3,691,841	4,245,967	5,869,365	ROA	19.2%	23.5%	30.4%	30.8%
LT investments	149,040	172,370	230,794	236,618	ROCE	22.7%	27.7%	35.0%	35.6%
Other non current assets	219,565	251,682	279,138	279,138	SG&A/Sales	3.4%	2.6%	2.2%	2.5%
Total assets	6,691,938	7,933,024	10,499,441	13,318,690	Net debt/equity	NM	NM	NM	NM
Short term borrowings	59,858	136,926	167,410	167,410	P/E (x)	53.6	36.6	22.4	17.2
Payables	74,227	84,330	115,239	129,548	P/BV (x)	14.7	11.6	8.2	6.0
Other short term liabilities	1,130,441	1,236,763	1,438,027	1,509,743	EV/EBITDA (x)	31.6	23.6	15.0	11.1
Current liabilities	1,264,525	1,458,019	1,720,676	1,806,701	Dividend Yield	0.7%	0.9%	1.2%	1.4%
Long-term debt	958,429	896,062	815,037	815,037	Sales/Assets (x)	47.4%	52.1%	59.5%	62.5%
Other long term liabilities	145,408	118,147	206,117	167,884	Interest cover (x)	NM	NM	NM	NM
Total liabilities	2,368,362	2,472,229	2,741,830	2,789,622	Operating leverage	128.3%	147.0%	149.7%	100.8%
Shareholders' equity	4,288,545	5,419,596	7,715,658	10,487,115	Revenue y/y Growth	33.9%	31.6%	43.8%	35.8%
Minority interests	35,031	41,199	41,953	41,953	EBITDA y/y Growth	36.5%	32.2%	53.0%	32.9%
Total liabilities & equity	6,691,938	7,933,024	10,499,441	13,318,690	Tax rate	16.6%	16.0%	17.2%	17.7%
BVPS	165.4	209.0	297.5	404.4	Adj. Net Income y/y Growth	39.9%	46.4%	63.3%	30.7%
y/y Growth	24.0%	26.4%	42.4%	35.9%	EPS y/y Growth	39.9%	46.4%	63.3%	30.7%
Net debt/(cash)	(1,403,733)	(2,035,607)	(3,730,838)	(4,653,157)	DPS y/y Growth	26.9%	33.3%	31.8%	19.0%
Cash flow from operating activities	1,975,661	2,437,716	3,564,030	4,453,920					
o/w Depreciation & amortization	662,797	688,096	808,126	968,602					
o/w Changes in working capital	(90,016)	(108)	(285,377)	(216,032)					
Cash flow from investing activities	(928,044)	(1,200,404)	(1,448,132)	(2,597,824)					
o/w Capital expenditure	(958,198)	(1,272,219)	(1,972,852)	(2,592,000)					
as % of sales	33.1%	33.4%	36.0%	34.8%					
Cash flow from financing activities	(313,242)	(590,737)	(471,208)	(933,777)					
o/w Dividends paid	(428,731)	(573,004)	(751,062)	(893,792)					
o/w Net debt issued/(repaid)	90,711	14,701	(50,541)	0					
Net change in cash	734,375	646,575	1,644,690	922,319					
Adj. Free cash flow to firm	1,142,360	1,292,759	2,201,778	1,861,920					
y/y Growth	164.7%	13.2%	70.3%	(15.4%)					

Source: Company reports and J.P. Morgan estimates.

Note: NT\$ in millions (except per-share data). Fiscal year ends Dec. o/w - out of which

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TSMC (2330.TW, 2330 TT) Price Chart



Date	Rating	Price (NT\$)	Price Target (NT\$)
20-Oct-23	OW	546.0	700
04-Jan-24	OW	578.0	750
19-Jan-24	OW	588.0	770
06-Mar-24	OW	730.0	850
07-Apr-24	OW	780.0	900
17-Jun-24	OW	922.0	1,080
19-Jul-24	OW	1005.0	1,200
18-Oct-24	OW	1035.0	1,500
13-Apr-25	OW	889.0	1,300
18-Apr-25	OW	847.0	1,275
24-Sep-25	OW	1340.0	1,550
17-Oct-25	OW	1485.0	1,700
07-Jan-26	OW	1705.0	2,100
16-Jan-26	OW	1690.0	2,250
02-Apr-26	OW	1855.0	2,400
17-Apr-26	OW	2085.0	2,500
07-Jul-26	OW	2460.0	3,100
17-Jul-26	OW	2470.0	3,200

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 28, 2002. All share prices are as of market close on the previous business day.

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